

32X COMM Registers (SH2 ↔ Sub-CPU)

The SH2 and Sub-CPU communicate via the following 32X COMM registers:

- **COMM0** (0x20004020 / 0xA15120): Carries the command number from the SH2 in the low byte. For CMD 48, the high byte carries a sound-effect ID. The Sub-CPU clears this to 0 when the command finishes.
- **COMM2** (0x20004022 / 0xA15122): Carries the command parameter, such as file number, track number, chunk index, or word index.
- **COMM4** (0x20004024 / 0xA15124): Exclusively used during start-up to transmit "S_OK" and the legacy 0x1234 signal.
- **COMM6** (0x20004026 / 0xA15126): Holds the Genesis vblank counter, acting as a diagnostic heartbeat (V).
- **COMM8 / COMM10** (0x20004028 / 0x2000402A | 0xA15128 / 0xA1512A): Used to transmit replies. For CMD 48, COMM8 returns the pad state and COMM10 returns the Genesis tick counter. It is also used to transmit file lengths and the 0x0AAA / 0x0BBB start-up handshake.
- **COMM12** (0x2000402C / 0xA1512C): Transmits the current SH2 status code (STATE) and 16-bit command replies, such as cart chunk results, sound results, and backup-RAM results.
- **COMM14** (0x2000402E / 0xA1512E): Transmits the Sub-CPU heartbeat (P), combining the CD drive-status nibble and the pass counter.

Sega CD Comm Registers (Sub-CPU ↔ Genesis 68000)

The Sub-CPU coordinates with the Genesis 68000 using the following registers:

- 0xFF8020 ... 0xFF802F: Receives up to four long arguments sent by the Sub-CPU.
- 0xFF800F: The Sub-CPU's command flag where it writes the command number.
- 0xFF800E: The Genesis flag where it echoes the command number to acknowledge receipt.
- 0xFF8010: The register where the Sub-CPU reads the returned result.
- 0xA1200F: The Sub flag on the Genesis side, which must be read twice to avoid torn read exceptions.

SH2 Commands (CMD)

These commands are sent by the SH2 in sh2_main.c and handled by the Sub-CPU:

- **38 (Open File)**: Uses COMM2 for the file number and returns the length in COMM8.
- **45 (Play CD Track)**: Uses COMM2 for the track number and triggers BIOS play-repeat.
- **46 (Chunk → Cart)**: Used in Cart mode for verified chunk writes.
- **48 (Pad)**: Replies with the pad and ticks in COMM8/10, while the high byte of COMM0 starts a sound effect.
- **50 (File Chunk → Frame Buffer)**: Used in No-Cart mode to transfer 16 KB off-screen.
- **51 (File Chunk → Frame Buffer)**: Used to load the 32 KB Splash screen at an offset.
- **52 (Genesis Display On/Off)**: Toggles the Genesis text layer.
- **53 (Load Sound Pack)**: Loads the sound pack into the PCM chip.
- **54 (Stop Music)**: Triggers the BIOS MSCSTOP function.

- 56 (**Backup RAM Load**): Loads the backup RAM file.
- 57 (**Backup RAM Get**): Retrieves a backup RAM word.
- 58 (**Backup RAM Put**): Writes a backup RAM word.
- 59 (**Backup RAM Save**): Saves the backup RAM file.

Genesis 68000 Commands (MD_CMD)

These commands are sent by the Sub-CPU and executed by the Genesis 68000:

- 1: MD_CMD_INIT_HW
- 2: MD_CMD_SET_SR (Disabled due to torn read hazards)
- 3: MD_CMD_GET_PAD
- 4: MD_CMD_CLEAR_B
- 5: MD_CMD_SET_VRAM
- 6: MD_CMD_NEXT_VRAM
- 7: MD_CMD_COPY_VRAM
- 8: MD_CMD_CLEAR_A
- 9: MD_CMD_PUT_STR
- 10: MD_CMD_PUT_CHR
- 11: MD_CMD_DELAY
- 12: MD_CMD_SET_PALETTE
- 13: MD_CMD_Z80_BUSREQUEST
- 14: MD_CMD_Z80_RESET
- 15: MD_CMD_Z80_MEMCLR
- 16: MD_CMD_Z80_MEMCPY
- 17: MD_CMD_DMA_SCREEN
- 18: MD_CMD_INIT_32X
- 19: MD_CMD_READ_32X_16
- 20: MD_CMD_WRITE_32X_16
- 21: MD_CMD_READ_32X_32
- 22: MD_CMD_WRITE_32X_32
- 23: MD_CMD_COPY_TO_CART
- 24: MD_CMD_COPY_FROM_CART

- 25: MD_CMD_COPY_WORDS
- 26: MD_CMD_SAFE_CART_XFER
- 27: MD_CMD_SAFE_CART_XFER_W
- 28: MD_CMD_BRAM_PUT
- 29: MD_CMD_BRAM_TO_FB
- 30: MD_CMD_BRAM_GET
- 31: MD_CMD_INSTALL_EXC (Hooks the 68K exception trap)
- 32: MD_CMD_CHILLY_MEMCPY
- 33: MD_CMD_CHILLY_RV_MEMCPY
- 34: MD_CMD_SET_VDP_REG
- 35: MD_CMD_END

Diagnostic Readout Lines

These indicators are displayed on the right-hand diagnostic strip to track system status:

- T: Tile set 1 offset from the bank table, comparing the frame buffer value against the SH2 copy.
- S: Star offset from the bank table, comparing the frame buffer value against the SH2 copy.
- 1/2/3: Displays the replies for CMD 50, showing chunk, sectors, and the first data word.
- N: The total number of CMD 50 requests sent.
- X/Y/D: The player ship's world X position, Y position, and current direction.
- B: The number of bolts currently in flight.
- E: The number of enemies currently alive.
- L: The player's remaining lives.
- G: The number of active cannons/guns.
- P: The Sub-CPU heartbeat from COMM14. The top digit represents the CD drive status register (0xFF8038) where: 0 = stopped, 1 = playing, 2 = seeking, 3 = scanning, 4 = paused, 5 = tray open, 9 = reading TOC, and B = no disc / error. The lower three digits represent the pass count divided by 64.

SH2 Status Codes (STATE / COMM12)

These hex codes map to the current boot and runtime phase of the SH2 engine:

- 4B01 (ST_LOADING): Load has started. CMD 38 opens the graphics file and CMD 46 copies chunks to the cart.
- 4B02 (ST_PACK_OK): Pack successfully loaded into the cart.
- 4B03: Graphics have been copied from the cart into SDRAM.

- 4B04: The sound pack is loaded into the PCM chip via CMD 53.
- 4B05: High scores and options are read from backup RAM via CMD 56/57.
- 4B51 – 4B54: Palette loaded, level 1 built, 256-colour mode and text set via CMD 52, and buffer swaps completed.
- 4B6n / 4B7n: Tile set *n* is copied from the cart or written into the frame buffers.
- 4B06 / 4B07: Tile sets are finished and the panel is drawn.
- 4B08: Music started via CMD 45. No CD data reads occur after this point.
- 4B10 (ST_RUNNING): The main loop is actively running the menu or game.
- 4BE1 (ST_ERR_OPEN): KOBOGFX.BIN was not found on the CD.
- 4BE2 (ST_ERR_VERIFY): A chunk failed verification in the cart.
- 4BE3 (ST_ERR_MAGIC): The cart data does not start with the required 'KGFX' signature.
- 4BE5: The pack header banks exceed the maximum allowed limit.
- 4BE6: The program is too big and exceeds the 0x0600C000 memory boundary.

Genesis Hardware Error Codes

These codes are returned during the init_32x phase if the 32X adapter fails to initialize:

- -2: No 32X detected (the Genesis did not find the "MARS" signature at 0xA130EC).
- -3: SDRAM error (the 32X boot ROM self-test failed and returned "SDER").
- -5: Adapter not ready timeout (timed out waiting for bit 7 of 0xA15101 to clear).